

ShipmentInfo

```
public class ShipmentInfo {  
    private String shipmentId;  
    private String packageType;  
    private int ecoPriorityScore;  
    private String dispatchDate;  
    private double estimatedTransitTime;  
    private double carbonTaxCost;  
    private String origin;  
    private String destination;  
  
    public ShipmentInfo(String shipmentId, String packageType, int  
ecoPriorityScore,  
    String dispatchDate, double estimatedTransitTime, double  
carbonTaxCost,  
    String origin, String destination) {  
        this.shipmentId = shipmentId;  
        this.packageType = packageType;  
        this.ecoPriorityScore = ecoPriorityScore;  
        this.dispatchDate = dispatchDate;  
        this.estimatedTransitTime = estimatedTransitTime;  
        this.carbonTaxCost = carbonTaxCost;  
        this.origin = origin;  
        this.destination = destination;  
    }  
  
    public String getShipmentId() { return shipmentId; }  
  
    public String getPackageType() { return packageType; }  
  
    public int getEcoPriorityScore() { return ecoPriorityScore; }  
}
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    public String getDispatchDate() { return dispatchDate; }

    public double getEstimatedTransitTime() { return estimatedTransitTime;
}

    public double getCarbonTaxCost() { return carbonTaxCost; }

    public String getOrigin() { return origin; }

    public String getDestination() { return destination; }

    @Override
    public String toString() {
        return shipmentId + " - " + packageType;
    }
}

```

CarrierInfo

```

import java.util.*;

public class CarrierInfo {
    private String carrierId;
    private String carrierName;
    private String fleetType;
    private List<ShipmentInfo> shipments; // Has-A Relationship

    public CarrierInfo(String carrierId, String carrierName, String
fleetType) {
        this.carrierId = carrierId;
        this.carrierName = carrierName;
    }
}

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        this.fleetType = fleetType;
        this.shipments = new ArrayList<>();
    }

    public void addShipment(ShipmentInfo shipment) {
        this.shipments.add(shipment);
    }

    public double getTotalCarbonTaxCost() {
        double total = 0;
        for (ShipmentInfo s : shipments) {
            total += s.getCarbonTaxCost();
        }
        return total;
    }

    public String getCarrierId() { return carrierId; }

    public String getCarrierName() { return carrierName; }

    public String getFleetType() { return fleetType; }

    public List<ShipmentInfo> getShipments() {
        return new ArrayList<>(shipments);
    }
}

```

LogisticsHubGUI

```

import java.util.*;
import java.io.*;

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import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.awt.*;

public class LogisticsHubGUI extends JFrame {

    // Structural collections mapping out data paths
    private LinkedList<CarrierInfo> carrierList = new LinkedList<>();
    private Queue<CarrierInfo> queue1 = new ArrayDeque<>();
    private Queue<CarrierInfo> queue2 = new ArrayDeque<>();
    private Queue<CarrierInfo> queue3 = new ArrayDeque<>();
    private Stack<CarrierInfo> dispatchedStack = new Stack<>();

    // UI Tables
    private JTable tblMasterData, tblQueue1, tblQueue2, tblQueue3,
tblStack;

    private DefaultTableModel modelMaster, modelQ1, modelQ2, modelQ3,
modelStack;

    // Interactive Input Fields
    private JTextField txtCarrierId, txtCarrierName, txtShipmentId,
txtPackageType, txtDate, txtTime, txtTax, txtOrigin, txtDest;
    private JComboBox<String> cmbFleetType, cmbEcoScore;
    private JButton btnAddManual, btnLoadFile, btnRoute, btnProcess;
    private JLabel lblStatus;

    public LogisticsHubGUI() {
        super("UPTM Smart Logistics Sustainable Workspace Hub");
        initializeGUI();
    }

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private void initializeGUI() {
    setSize(1350, 800);
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    setLocationRelativeTo(null);
    setLayout(new BorderLayout(10, 10));

    // 1. TOP PANEL: Title & Operational Pipeline Controls
    JPanel pnlTop = new JPanel(new FlowLayout(FlowLayout.LEFT, 15,
10));
    pnlTop.setBackground(new Color(41, 128, 185));

    btnLoadFile = new JButton("1. Auto-Load File (>=100 Records)");
    btnRoute = new JButton("2. Execute Conveyor Routing");
    btnProcess = new JButton("3. Run Pipeline Batches");

    lblStatus = new JLabel("System Status: Awaiting Action");
    lblStatus.setForeground(Color.WHITE);
    lblStatus.setFont(new Font("Arial", Font.BOLD, 12));

    pnlTop.add(btnLoadFile);
    pnlTop.add(btnRoute);
    pnlTop.add(btnProcess);
    pnlTop.add(lblStatus);

    // 2. LEFT PANEL: Interactive Manual Entry Form
    JPanel pnlInteractiveForm = new JPanel(new GridBagLayout());

    pnlInteractiveForm.setBorder(BorderFactory.createTitledBorder("Interactive
Manual Data Entry Panel"));

    pnlInteractiveForm.setPreferredSize(new Dimension(360, 800));
    GridBagConstraints gbc = new GridBagConstraints();

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gbc.insets = new Insets(4, 6, 4, 6);
gbc.fill = GridBagConstraints.HORIZONTAL;
gbc.anchor = GridBagConstraints.WEST;

// Form Fields Initialization
txtCarrierId = new JTextField("CR999", 12);
txtCarrierName = new JTextField("Pos Malaysia Green", 12);
cmbFleetType = new JComboBox<>(new String[]{"Electric Van", "Drone
Delivery", "Hybrid Truck", "Freight Train", "LNG Cargo Vessel", "Biofuel
Air Freight"});

txtShipmentId = new JTextField("SHP777", 12);
txtPackageType = new JTextField("Electronics Bulk", 12);
cmbEcoScore = new JComboBox<>(new String[]{"1", "2", "3", "4",
"5"});

cmbEcoScore.setSelectedIndex(4);
txtDate = new JTextField("2026-06-25", 12);
txtTime = new JTextField("24.0", 12);
txtTax = new JTextField("350.00", 12);
txtOrigin = new JTextField("Port Klang", 12);
txtDest = new JTextField("Kuching", 12);
btnAddManual = new JButton("Add Record to System");

// UI Placement Assembly Loop
Component[] comps = {
    new JLabel("Carrier ID:"), txtCarrierId,
    new JLabel("Carrier Name:"), txtCarrierName,
    new JLabel("Fleet Type:"), cmbFleetType,
    new JLabel("Shipment ID:"), txtShipmentId,
    new JLabel("Package Type:"), txtPackageType,
    new JLabel("Eco Priority Score (1-5):"), cmbEcoScore,
    new JLabel("Dispatch Date:"), txtDate,

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        new JLabel("Transit Time (Hours):"), txtTime,
        new JLabel("Carbon Tax Cost (RM):"), txtTax,
        new JLabel("Origin Location:"), txtOrigin,
        new JLabel("Destination Location:"), txtDest
    };

    int r = 0;
    for (int i = 0; i < comps.length; i += 2) {
        gbc.gridx = 0; gbc.gridy = r; pnlInteractiveForm.add(comps[i],
gbc);

        gbc.gridx = 1; pnlInteractiveForm.add(comps[i+1], gbc);

        r++;
    }

    gbc.gridx = 0; gbc.gridy = r; gbc.gridwidth = 2;
    gbc.insets = new Insets(15, 5, 5, 5);
    pnlInteractiveForm.add(btnAddManual, gbc);

    // 3. CENTER PANEL: Outputs Display Tabs
    JTabbedPane tabbedPane = new JTabbedPane();

    modelMaster = new DefaultTableModel(new String[]{"Carrier ID",
    "Name", "Fleet Class", "Shipment ID", "Route From -> To", "Tax"}, 0);
    tblMasterData = new JTable(modelMaster);
    tabbedPane.addTab("Phase 1: Master LinkedList View", new
    JScrollPane(tblMasterData));

    JPanel pnlQueues = new JPanel(new GridLayout(3, 1, 5, 5));
    modelQ1 = new DefaultTableModel(new String[]{"Carrier Name",
    "Fleet Type", "Shipments Count", "Accumulated Tax"}, 0);
    tblQueue1 = new JTable(modelQ1);
    JPanel pnlQ1 = new JPanel(new BorderLayout());

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        pnlQ1.setBorder(BorderFactory.createTitledBorder("LANE 1: Regional
Micro-Distribution Fleet Queue"));

        pnlQ1.add(new JScrollPane(tblQueue1), BorderLayout.CENTER);

        modelQ2 = new DefaultTableModel(new String[]{"Carrier Name",
"Fleet Type", "Shipments Count", "Accumulated Tax"}, 0);

        tblQueue2 = new JTable(modelQ2);

        JPanel pnlQ2 = new JPanel(new BorderLayout());

        pnlQ2.setBorder(BorderFactory.createTitledBorder("LANE 2: Cross -
Border Express Logistics Queue"));

        pnlQ2.add(new JScrollPane(tblQueue2), BorderLayout.CENTER);

        modelQ3 = new DefaultTableModel(new String[]{"Carrier Name",
"Fleet Type", "Shipments Count", "Accumulated Tax"}, 0);

        tblQueue3 = new JTable(modelQ3);

        JPanel pnlQ3 = new JPanel(new BorderLayout());

        pnlQ3.setBorder(BorderFactory.createTitledBorder("LANE 3:
Industrial Heavy Bulk Haulage Queue"));

        pnlQ3.add(new JScrollPane(tblQueue3), BorderLayout.CENTER);

        pnlQueues.add(pnlQ1); pnlQueues.add(pnlQ2); pnlQueues.add(pnlQ3);
        tabbedPane.addTab("Phase 2: Queue Conveyor Lanes", pnlQueues);

        modelStack = new DefaultTableModel(new String[]{"Cleared Carrier
Name", "Assigned Fleet System", "Total Loaded Items", "Settled Carbon
Tax"}, 0);

        tblStack = new JTable(modelStack);

        tabbedPane.addTab("Phase 3: DispatchedStack Departure Log", new
JScrollPane(tblStack));

        add(pnlTop, BorderLayout.NORTH);

        add(pnlInteractiveForm, BorderLayout.WEST);

        add(tabbedPane, BorderLayout.CENTER);

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// Action Listeners Execution Wire-up
btnAddManual.addActionListener(e -> addManualRecordAction());
btnLoadFile.addActionListener(e -> loadManifestFileAction());
btnRoute.addActionListener(e -> routeConveyorAction());
btnProcess.addActionListener(e -> processPipelineAction());
}

private void addManualRecordAction() {
    try {
        String cId = txtCarrierId.getText().trim();
        String cName = txtCarrierName.getText().trim();
        String fType = cmbFleetType.getSelectedItem().toString();
        String sId = txtShipmentId.getText().trim();
        String pType = txtPackageType.getText().trim();
        int ecoScore =
Integer.parseInt(cmbEcoScore.getSelectedItem().toString());
        String date = txtDate.getText().trim();
        double time = Double.parseDouble(txtTime.getText().trim());
        double tax = Double.parseDouble(txtTax.getText().trim());
        String origin = txtOrigin.getText().trim();
        String dest = txtDest.getText().trim();

        if (cId.isEmpty() || cName.isEmpty() || sId.isEmpty()) {
            JOptionPane.showMessageDialog(this, "Please fill in all
core identifier data fields.", "Warning", JOptionPane.WARNING_MESSAGE);
            return;
        }

        ShipmentInfo shipment = new ShipmentInfo(sId, pType, ecoScore,
date, time, tax, origin, dest);

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        CarrierInfo carrier = null;
        for (CarrierInfo c : carrierList) {
            if (c.getCarrierId().equals(cId)) {
                carrier = c;
                break;
            }
        }
        if (carrier == null) {
            carrier = new CarrierInfo(cId, cName, fType);
            carrierList.add(carrier);
        }
        carrier.addShipment(shipment);

        modelMaster.addRow(new Object[]{cId, cName, fType, sId, origin
+ " -> " + dest, "RM " + tax});

        lblStatus.setText("Status: Manual Entry added (" + sId + ")");

        JOptionPane.showMessageDialog(this, "Successfully combined
record into the Master LinkedList.");

    } catch (NumberFormatException ex) {

        JOptionPane.showMessageDialog(this, "Please enter valid
numbers for score, time, and tax data items.", "Input Error",
JOptionPane.ERROR_MESSAGE);

    }

}

private void loadManifestFileAction() {
    carrierList.clear();

    modelMaster.setRowCount(0);

    String filename = "supply_chain_manifest.txt";

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        try (BufferedReader br = new BufferedReader(new
FileReader(filename))) {
            String line;
            int recordCounter = 0;
            while ((line = br.readLine()) != null) {
                if (line.trim().isEmpty()) continue;
                String[] tokens = line.split(",");
                if (tokens.length < 11) continue;

                String cId = tokens[0].trim();
                String cName = tokens[1].trim();
                String fType = tokens[2].trim();
                String sId = tokens[3].trim();
                String pType = tokens[4].trim();
                int ecoScore = Integer.parseInt(tokens[5].trim());
                String date = tokens[6].trim();
                double time = Double.parseDouble(tokens[7].trim());
                double tax = Double.parseDouble(tokens[8].trim());
                String origin = tokens[9].trim();
                String dest = tokens[10].trim();

                ShipmentInfo shipment = new ShipmentInfo(sId, pType,
ecoScore, date, time, tax, origin, dest);

                CarrierInfo carrier = null;
                for (CarrierInfo c : carrierList) {
                    if (c.getCarrierId().equals(cId)) {
                        carrier = c;
                        break;
                    }
                }
            }
        }

```

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        if (carrier == null) {
            carrier = new CarrierInfo(cId, cName, fType);
            carrierList.add(carrier);
        }
        carrier.addShipment(shipment);

        modelMaster.addRow(new Object[]{cId, cName, fType, sId,
origin + " -> " + dest, "RM " + tax});
        recordCounter++;
    }

    lblStatus.setText("Status: Loaded! Parsed Records: " +
recordCounter);

    JOptionPane.showMessageDialog(this, "Successfully compiled " +
recordCounter + " text rows into Master LinkedList.");
    } catch (IOException | NumberFormatException ex) {
        lblStatus.setText("Status: Error Reading File!");

        JOptionPane.showMessageDialog(this, "Make sure
supply_chain_manifest.txt exists in the source folder.", "Error",
JOptionPane.ERROR_MESSAGE);
    }
}

private void routeConveyorAction() {
    if (carrierList.isEmpty()) {
        JOptionPane.showMessageDialog(this, "Please enter or load data
first.");
        return;
    }

    queue1.clear(); queue2.clear(); queue3.clear();

    modelQ1.setRowCount(0); modelQ2.setRowCount(0);
modelQ3.setRowCount(0);

    boolean alternateToLane1 = true;

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        for (CarrierInfo carrier : carrierList) {
            int shipmentVolume = carrier.getShipments().size();

            if (shipmentVolume > 3) {
                queue3.add(carrier);

                modelQ3.addRow(new Object[]{carrier.getCarrierName(),
carrier.getFleetType(), shipmentVolume, "RM " +
carrier.getTotalCarbonTaxCost()});
            } else {
                if (alternateToLane1) {
                    queue1.add(carrier);

                    modelQ1.addRow(new Object[]{carrier.getCarrierName(),
carrier.getFleetType(), shipmentVolume, "RM " +
carrier.getTotalCarbonTaxCost()});
                } else {
                    queue2.add(carrier);

                    modelQ2.addRow(new Object[]{carrier.getCarrierName(),
carrier.getFleetType(), shipmentVolume, "RM " +
carrier.getTotalCarbonTaxCost()});
                }

                alternateToLane1 = !alternateToLane1;
            }
        }

        lblStatus.setText("Status: Conveyor Sorting Streams Allocated.");

        JOptionPane.showMessageDialog(this, "Queues populated!\nLane 1: "
+ queue1.size() + " | Lane 2: " + queue2.size() + " | Lane 3 (Bulk): " +
queue3.size());
    }

    private void processPipelineAction() {
        if (queue1.isEmpty() && queue2.isEmpty() && queue3.isEmpty()) {
            JOptionPane.showMessageDialog(this, "Lanes are empty. Run Step
2 first.");
        }
    }

```

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        return;
    }
    dispatchedStack.clear();
    modelStack.setRowCount(0);

    while (!queue1.isEmpty() || !queue2.isEmpty() ||
!queue3.isEmpty()) {
        int count1 = 0;
        while (!queue1.isEmpty() && count1 < 5) {
            dispatchedStack.push(queue1.poll());
            count1++;
        }
        int count2 = 0;
        while (!queue2.isEmpty() && count2 < 5) {
            dispatchedStack.push(queue2.poll());
            count2++;
        }
        int count3 = 0;
        while (!queue3.isEmpty() && count3 < 5) {
            dispatchedStack.push(queue3.poll());
            count3++;
        }
    }

    while (!dispatchedStack.isEmpty()) {
        CarrierInfo cleared = dispatchedStack.pop();
        modelStack.addRow(new Object[]{
            cleared.getCarrierName(),
            cleared.getFleetType(),
            cleared.getShipments().size(),
            String.format("RM %.2f", cleared.getTotalCarbonTaxCost())
        });
    }
}

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        });
    }

    lblStatus.setText("Status: Core Pipeline Manifests Dispatched via
Stack operations.");

    JOptionPane.showMessageDialog(this, "Dispatched Stack settlement
output logged successfully!");
}

public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> {
        try {

UIManager.setLookAndFeel(UIManager.getSystemLookAndFeelClassName());

            } catch (Exception ignored) {}

            new LogisticsHubGUI().setVisible(true);

        });
    }
}

```

GenerateManifest

```

import java.io.BufferedWriter;
import java.io.FileWriter;
import java.io.IOException;

public class GenerateManifest {

    public static void main(String[] args) {

        String filename = "supply_chain_manifest.txt";

        String data = "CR001,Pos Malaysia Green,Electric
Van,SHP101,Electronics,4,2026-06-01,2.5,15.50,Shah Alam,Kuala Lumpur\n" +

```

"CR001,Pos Malaysia Green,Electric Van,SHP102,Documents,5,2026-06-01,1.2,5.00,Shah Alam,Petaling Jaya\n" +

"CR001,Pos Malaysia Green,Electric Van,SHP103,Clothing,4,2026-06-02,3.0,12.00,Shah Alam,Klang\n" +

"CR001,Pos Malaysia Green,Electric Van,SHP104,Medical Kits,5,2026-06-02,2.0,20.00,Shah Alam,Subang Jaya\n" +

"CR002,DHL Express Asia,Drone Delivery,SHP201,Vaccines,5,2026-06-01,0.5,45.00,Penang Island,Bayan Lepas\n" +

"CR002,DHL Express Asia,Drone Delivery,SHP202,Microchips,5,2026-06-02,0.8,60.00,Penang Island,Seberang Perai\n" +

"CR003,KTM Cargo Eco,Freight Train,SHP301,Automotive Steel,3,2026-06-01,12.0,450.00,Ipoh,Port Klang\n" +

"CR003,KTM Cargo Eco,Freight Train,SHP302,Industrial Timber,3,2026-06-03,14.5,520.00,Alor Setar,Port Klang\n" +

"CR003,KTM Cargo Eco,Freight Train,SHP303,Mineral Ore,2,2026-06-04,18.0,680.00,Padang Besar,Port Klang\n" +

"CR003,KTM Cargo Eco,Freight Train,SHP304,Heavy Machinery,2,2026-06-05,10.5,410.00,Taiping,Shah Alam\n" +

"CR004,MISC Maritime,LNG Cargo Vessel,SHP401,Petroleum Sub,3,2026-06-01,48.0,2400.00,Bintulu,Port Klang\n" +

"CR004,MISC Maritime,LNG Cargo Vessel,SHP402,Liquefied Gas,3,2026-06-04,72.0,3120.00,Miri,Johor Port\n" +

"CR005,Malaysian EcoAir,Biofuel Air Freight,SHP501,Fresh Seafood,4,2026-06-02,1.8,850.00,Kota Kinabalu,Sepang\n" +

"CR005,Malaysian EcoAir,Biofuel Air Freight,SHP502,E-Commerce Bulk,3,2026-06-03,2.2,920.00,Kuching,Sepang\n" +

"CR006,City-Link Local,Hybrid Truck,SHP601,Groceries,4,2026-06-01,4.5,45.50,Melaka,Kuala Lumpur\n" +

"CR006,City-Link Local,Hybrid Truck,SHP602,Home Appliances,4,2026-06-02,5.0,75.00,Muar,Kuala Lumpur\n" +

"CR007,GDEX Regional,Electric Van,SHP701,Cosmetics,5,2026-06-01,3.2,18.00,Johor Bahru,Batu Pahat\n" +

"CR007,GDEX Regional,Electric Van,SHP702,Office Supplies,4,2026-06-02,2.8,14.00,Johor Bahru,Kluang\n" +

"CR008,NinjaVan Eco,Drone Delivery,SHP801,Lab Samples,5,2026-06-01,0.4,35.00,Kuala Lumpur,Cheras\n" +

"CR008,NinjaVan Eco,Drone Delivery,SHP802,Smartphones,5,2026-06-02,0.6,50.00,Kuala Lumpur,Ampang\n" +

"CR009,FedEx Borneo,Biofuel Air Freight,SHP901,Spare Parts,3,2026-06-02,1.5,410.00,Labuan,Miri\n" +

"CR009,FedEx Borneo,Biofuel Air Freight,SHP902,Pharmaceuticals,4,2026-06-04,2.0,580.00,Selangor,Sandakan\n" +

"CR010,J&T Green Fleet,Hybrid Truck,SHP001,Fast Fashion,4,2026-06-01,6.0,85.00,Kuantan,Kuala Lumpur\n" +

"CR010,J&T Green Fleet,Hybrid Truck,SHP002,Plastics Eco,3,2026-06-03,7.5,110.00,Kemaman,Kuala Lumpur\n" +

"CR011,Skyline Logistics,Electric Van,SH111,Art Prints,5,2026-06-05,1.5,9.50,Cyberjaya,Putrajaya\n" +

"CR011,Skyline Logistics,Electric Van,SH112,Books,5,2026-06-05,1.8,11.00,Cyberjaya,Serdang\n" +

"CR012,Oceanic Green,LNG Cargo Vessel,SH121,Scrap Metal,3,2026-06-06,36.0,1850.00,Sandakan,Port Klang\n" +

"CR012,Oceanic Green,LNG Cargo Vessel,SH122,Raw Rubber,4,2026-06-08,42.0,2100.00,Tawau,Johor Port\n" +

"CR013,Trans-Borneo,Hybrid Truck,SH131,Frozen Food,4,2026-06-05,12.0,230.00,Kuching,Sibu\n" +

"CR013,Trans-Borneo,Hybrid Truck,SH132,Construction Mat,3,2026-06-07,15.5,310.00,Kuching,Miri\n" +

"CR014,EcoMovers,Electric Van,SH141,Furniture Pack,4,2026-06-06,4.0,55.00,Rawang,Kuala Lumpur\n" +

"CR014,EcoMovers,Electric Van,SH142,Mattresses,4,2026-06-07,4.8,65.00,Sungei Buloh,Kuala Lumpur\n" +

"CR015,Alpha Courier,Drone Delivery,SH151,Blood Vials,5,2026-06-08,0.3,40.00,Seremban,Nilai\n" +

"CR015,Alpha Courier,Drone Delivery,SH152,Biomedical Sensors,5,2026-06-09,0.5,55.00,Seremban,Port Dickson\n" +

"CR016,MegaFreight,Freight Train,SH161,Cement Bags,2,2026-06-07,8.0,380.00,Gua Musang,Tumpat\n" +

"CR016,MegaFreight,Freight Train,SH162,Coal Clean,2,2026-06-09,11.5,490.00,Kuala Lipis,Kota Bharu\n" +

"CR017,Pacific BioAir,Biofuel Air Freight,SH171,Live Plants,4,2026-06-08,3.0,720.00,Penang,Kuching\n" +

"CR017,Pacific BioAir,Biofuel Air Freight,SH172,Lab Chemicals,4,2026-06-10,3.5,890.00,Selangor,Kota Kinabalu\n" +

"CR018,Express Green,Electric Van,SH181,Toys,5,2026-06-09,2.2,16.50,Klang,Subang\n" +

"CR018,Express Green,Electric Van,SH182,Footwear,4,2026-06-10,2.5,14.00,Klang,Shah Alam\n" +

"CR019,Borneo SeaHub,LNG Cargo Vessel,SH191,Palm Oil Bio,3,2026-06-10,24.0,1500.00,Lahad Datu,Bintulu\n" +

"CR019,Borneo SeaHub,LNG Cargo Vessel,SH192,Cocoa Beans,4,2026-06-12,30.0,1750.00,Tawau,Bintulu\n" +

"CR020,EastCoast Line,Hybrid Truck,SH201,Keropok Bulk,4,2026-06-10,8.0,95.00,Kuala Terengganu,Kuantan\n" +

"CR020,EastCoast Line,Hybrid Truck,SH202,Textiles,4,2026-06-11,9.0,120.00,Kota Bharu,Kuantan\n" +

"CR021,Utara EcoExpress,Electric Van,SH211,Hardware Tools,4,2026-06-12,3.0,22.00,Alor Setar,Sungai Petani\n" +

"CR021,Utara EcoExpress,Electric Van,SH212,Gardening Supplies,5,2026-06-13,2.5,19.00,Alor Setar,Kangar\n" +

"CR022,ZettaDrone,Drone Delivery,SH221,Insulin Packs,5,2026-06-11,0.6,42.00,Malacca Town,Jasin\n" +

"CR022,ZettaDrone,Drone Delivery,SH222,Tech Components,5,2026-06-12,0.7,48.00,Malacca Town,Alor Gajah\n" +

"CR023,Malaya RailEco,Freight Train,SH231,Fertilizer Eco,3,2026-06-12,15.0,610.00,Gemas,Johor Bahru\n" +

"CR023,Malaya RailEco,Freight Train,SH232,Paper Rolls,4,2026-06-14,13.0,540.00,Mentakab,Port Klang\n" +

"CR024,Global JetBio,Biofuel Air Freight,SH241,Luxury Goods,4,2026-06-13,1.2,980.00,Selangor,Penang\n" +

"CR024,Global JetBio,Biofuel Air Freight,SH242,Precision Lenses,5,2026-06-15,1.4,1150.00,Selangor,Senai\n" +

"CR025,Southern Hybrid,Hybrid Truck,SH251,Ceramic Tiles,3,2026-06-14,5.5,135.00,Kluang,Johor Bahru\n" +

"CR025,Southern Hybrid,Hybrid Truck,SH252,Glassware,4,2026-06-15,6.2,160.00,Yong Peng,Johor Bahru\n" +

"CR026,E-Link Express,Electric Van,SH261,Protein Powder,5,2026-06-15,2.0,15.00,Puchong,Kuala Lumpur\n" +

"CR026,E-Link Express,Electric Van,SH262,Sporting Goods,4,2026-06-16,2.4,18.50,Puchong,Petaling Jaya\n" +

"CR027,Vortex Shipping,LNG Cargo Vessel,SH271,Crude BioOil,3,2026-06-16,50.0,2800.00,Kemaman,Kerteh\n" +

"CR027,Vortex Shipping,LNG Cargo Vessel,SH272,Iron Rods,3,2026-06-19,65.0,3120.00,Kuantan Port,Singapore\n" +

"CR028,InterState Eco,Hybrid Truck,SH281,Grains,4,2026-06-17,10.0,190.00,Anson,Kuala Lumpur\n" +

"CR028,InterState Eco,Hybrid Truck,SH282,Canned Goods,4,2026-06-18,11.2,210.00,Taiping,Kuala Lumpur\n" +

"CR029,SwiftFlyer Drone,Drone Delivery,SH291,Antivenom,5,2026-06-16,0.2,65.00,Kangar,Arau\n" +

"CR029,SwiftFlyer Drone,Drone Delivery,SH292,Hard Drives,5,2026-06-17,0.4,55.00,Kangar,Padang Besar\n" +

"CR030,National EcoRail,Freight Train,SH301,Bauxite Ore,2,2026-06-18,16.0,730.00,Kuantan,Port Klang\n" +

"CR030,National EcoRail,Freight Train,SH302,Granite Blocks,2,2026-06-20,20.5,890.00,Ipoh,Johor Bahru\n" +

"CR031,EcoCargo Air,Biofuel Air Freight,SH311,Sea Food Cargo,4,2026-06-19,2.5,820.00,Tawau,Sepang\n" +

"CR031,EcoCargo Air,Biofuel Air Freight,SH312,Printed Circuit,5,2026-06-20,2.1,940.00,Kuching,Penang\n" +

"CR032,GreenGlide,Electric Van,SH321,Pet Supplies,5,2026-06-20,1.4,10.50,Kajang,Bangi\n" +

"CR032,GreenGlide,Electric Van,SH322,Snack Boxes,5,2026-06-21,1.1,8.00,Kajang,Semenyih\n" +

"CR033,Meridian Line,LNG Cargo Vessel,SH331,Wheat Bulk,4,2026-06-21,72.0,3120.00,Port Klang,Labuan\n" +

"CR033,Meridian Line,LNG Cargo Vessel,SH332,Soybeans,4,2026-06-24,80.0,3120.00,Port Klang,Kuching\n" +

"CR034,Capital Movers,Hybrid Truck,SH341,Office Tables,4,2026-06-22,3.5,65.00,Semenyih,Putrajaya\n" +

"CR034,Capital Movers,Hybrid Truck,SH342,Filing Cabinets,4,2026-06-23,4.0,70.00,Semenyih,Cyberjaya\n" +

"CR035,Hexa Drone,Drone Delivery,SH351,First Aid Packs,5,2026-06-22,0.5,38.00,Sibu,Kapit\n" +

"CR035,Hexa Drone,Drone Delivery,SH352,Emergency Radio,5,2026-06-23,0.6,44.00,Sibu,Kanowit\n" +

"CR036,Central Freight,Freight Train,SH361,Gypsum Boards,3,2026-06-23,9.5,410.00,Batu Gajah,Port Klang\n" +

"CR036,Central Freight,Freight Train,SH362,Brick Pallets,3,2026-06-25,11.0,460.00,Rawang,Batu Pahat\n" +

"CR037,AeroGreen Cargo,Biofuel Air Freight,SH371,Engine Sensors,5,2026-06-24,1.9,870.00,Senai,Sepang\n" +

"CR037,AeroGreen Cargo,Biofuel Air Freight,SH372,Vitamins Bulk,4,2026-06-26,2.3,790.00,Sepang,Miri\n" +

"CR038,SmartVan Trans,Electric Van,SH381,Teas & Coffees,5,2026-06-25,2.6,19.50,Gombak,Kuala Lumpur\n" +

"CR038,SmartVan Trans,Electric Van,SH382,Confectionery,5,2026-06-26,2.1,15.00,Gombak,Petaling Jaya\n" +

"CR039,Horizon Shipping,LNG Cargo Vessel,SH391,Bio Ethanol,3,2026-06-25,55.0,2950.00,Bintulu,Johor Port\n" +

"CR039,Horizon Shipping,LNG Cargo Vessel,SH392,Copper Wires,4,2026-06-28,60.0,3120.00,Port Klang,Kota Kinabalu\n" +

"CR040,CrossBorder Hyb,Hybrid Truck,SH401,Detergent Eco,4,2026-06-26,14.0,280.00,Bukit Kayu Hitam,Ipoh\n" +

"CR040,CrossBorder Hyb,Hybrid Truck,SH402,Solar Inverters,5,2026-06-27,16.5,390.00,Shah Alam,Padang Besar\n" +

"CR041,EcoWeave Post,Electric Van,SH411,Organic Towels,5,2026-06-26,1.7,13.00,Nilai,Seremban\n" +

"CR041,EcoWeave Post,Electric Van,SH412,Bedding Sets,4,2026-06-27,2.1,16.50,Nilai,Sepang\n" +

"CR042,Falcon Aero,Drone Delivery,SH421,Cardiac Monitor,5,2026-06-26,0.3,70.00,Kota Bharu,Pasir Mas\n" +

"CR042,Falcon Aero,Drone Delivery,SH422,Drone Battery,5,2026-06-27,0.4,45.00,Kota Bharu,Kubang Kerian\n" +

"CR043,IronTrack Eco,Freight Train,SH431,Steel Coils,2,2026-06-26,13.5,580.00,Pasir Gudang,Port Klang\n" +

"CR043,IronTrack Eco,Freight Train,SH432,Aluminium Bars,3,2026-06-28,15.0,620.00,Gemas,Port Klang\n" +

"CR044,Starlight Air,Biofuel Air Freight,SH441,Server Racks,4,2026-06-27,2.8,1100.00,Sepang,Kuching\n" +

"CR044,Starlight Air,Biofuel Air Freight,SH442,Optic Modules,5,2026-06-29,2.0,950.00,Penang,Senai\n" +

"CR045,Peninsular Hyb,Hybrid Truck,SH451,Mineral Water,4,2026-06-27,7.0,110.00,Taiping,Shah Alam\n" +

"CR045,Peninsular Hyb,Hybrid Truck,SH452,Packaging Box,4,2026-06-28,6.5,95.00,Sitiawan,Kuala Lumpur\n" +

"CR046,DirectVan Green,Electric Van,SH461,Baby Clothes,5,2026-06-27,1.9,14.00,Ampang,Kuala Lumpur\n" +

"CR046,DirectVan Green,Electric Van,SH462,Skincare Serum,5,2026-06-28,1.2,10.00,Ampang,Petaling Jaya\n" +

"CR047,Titan Bulk Sea,LNG Cargo Vessel,SH471,Liquefied Argon,3,2026-06-27,90.0,3120.00,Kerteh,Port Klang\n" +

"CR047,Titan Bulk Sea,LNG Cargo Vessel,SH472,Molten Sulfur,2,2026-06-30,110.0,3120.00,Kemaman,Johor Port\n" +

"CR048,Island Connect,Hybrid Truck,SH481,Rice Sacks,4,2026-06-28,4.5,75.00,Kangar,Butterworth\n" +

"CR048,Island Connect,Hybrid Truck,SH482,Cooking Oil,4,2026-06-29,5.2,85.00,Alor Setar,Butterworth\n" +

"CR049,Quantum Drone,Drone Delivery,SH491,Plasma Packs,5,2026-06-28,0.4,55.00,Kuala Terengganu,Marang\n" +

"CR049,Quantum Drone,Drone Delivery,SH492,Lab Glassware,5,2026-06-29,0.5,40.00,Kuala Terengganu,Gong Badak\n" +

"CR050,InterIsland Rail,Freight Train,SH501,Plywood Crate,3,2026-06-29,22.0,920.00,Mentakab,Tumpat\n" +

"CR050,InterIsland Rail,Freight Train,SH502,Rubber Sheets,4,2026-06-30,19.0,840.00,Gua Musang,Port Klang";

```
        try (BufferedWriter writer = new BufferedWriter(new
FileWriter(filename))) {

            writer.write(data);

            System.out.println("Success! 'supply_chain_manifest.txt' has
been generated in your working directory.");

        } catch (IOException e) {

            System.out.println("An error occurred while creating the file:
" + e.getMessage());

        }
```

}

}